

WEEK 5 - POWER BI EDA REPORT

Healthcare Data Cleaning, Transformation and Data Modeling

INTRODUCTION:

Microsoft Power BI is a business intelligence and data visualization tool used to clean, transform, organize and analyze data.

In this Week 5 activity, I worked with a healthcare dataset using Microsoft Power BI. The dataset contains information related to patients, hospital admissions, medical conditions and billing.

I used Power Query Editor to clean and transform the raw healthcare data. Different data transformation techniques were applied to improve the quality and consistency of the dataset. An Age Group column was also created using the Conditional Column feature.

After completing the data transformation, the dataset was organized into three separate tables: Dim_Patient, Dim_Admission and Fact_Billing. Relationships were then created between the tables to form a structured data model.

This activity helped me understand Power Query, data cleaning, data transformation, conditional columns, data modeling and relationships in Power BI.

OBJECTIVE:

The main objective of this Week 5 activity is to clean, transform and organize the healthcare dataset using Microsoft Power BI.

The objectives are:

- To import the healthcare dataset into Power BI.
- To clean and transform the raw healthcare data.
- To correct the data types of different columns.
- To clean and standardize text values.
- To remove errors and duplicate records.
- To remove unnecessary columns.
- To create an Age Group column using Conditional Column.
- To divide the dataset into meaningful tables.
- To create relationships between the tables.

DATASET:

For this activity, I used a healthcare dataset in Microsoft Power BI.

The dataset contains information related to patients, hospital admissions, medical conditions and billing.

The major fields in the dataset include:

- Patient ID
- Name
- Age
- Gender
- Blood Type

- Medical Condition
- Doctor
- Hospital
- Insurance Provider
- Admission Type
- Date of Admission
- Discharge Date
- Medication
- Test Results
- Billing Amount
- Room Number
- Admission ID

The dataset was used for data cleaning, transformation and data modeling in Power BI.

Name	Age	Gender	Blood Type	Medical Condition	Date of Admission	Doctor	Hospital
Kristen Flores	64	Male	A+	Arthritis	11 January 2021	Angela Reed	Carson and Nielsen, Buchanan
Eric Henderson	74	Female	A-	Asthma	08 March 2020	Allison Stone	and Wright, Coffey Eaton
Samantha Nguyen	70	Female	AB+	Diabetes	23 November 2019	Wayne Morales	Ward, and Clarke Ayala
Johnny Evans	70	Female	A-	Asthma	01 August 2020	Arthur Greene	Inc Adkins
Mr. James Mitchell	69	Male	O+	Arthritis	08 October 2023	Cindy Cooper	Guerrero LLC
Joshua Macdonald	73	Female	A-	Cancer	10 March 2021	Micheal Bradley	Cooper and Arnold Hernandez,
Jeffrey Fitzpatrick	57	Female	O-	Hypertension	17 October 2020	Victoria Allen	Inc Taylor
Carolyn Gordon	74	Female	A+	Cancer	07 September 2019	Michael Evans	Davis Bishop, Thomas and
Victoria Chambers	64	Male	A+	Hypertension	30 May 2022	Jordan Johnson	Sons Wilkerson and
Alexandra Phillips	62	Male	A+	Diabetes	03 April 2021	Michelle Williamson	Hutchinson-Haney
Jeff Strong	71	Male	A-	Hypertension	12 March 2021	Shane Taylor	Group Savage
Paige Ward	67	Female	O-	Cancer	24 August 2023	Jane Berry	Ewing Matthews Pham, and
Jennifer Moore	57	Female	A+	Obesity	12 August 2021	Erik Adams	and Ellis, Williamson Thomas
David Rivera	62	Male	O-	Asthma	30 March 2022	Leah Reed	Simmons PLC
Caitlyn Bright	71	Female	A-	Asthma	22 April 2023	Kathleen Perez	Group Delgado
Daniel Mendoza	70	Male	A+	Obesity	07 October 2020	Dr. Brandy Blair	Hansen-Howard
Michael Brown	71	Male	B-	Hypertension	04 April 2022	Nicole Alvarez	Peters-Greene
Dr. Keith Flynn	56	Male	A+	Cancer	27 April 2023	Dr. Mary Santiago DDS	Torres-Cardenas

Table: healthcare_dataset (55,500 rows)

POWER QUERY DATA TRANSFORMATION:

Power Query Editor was used to prepare the raw healthcare dataset before creating the final data model.

The following transformation steps were performed.

Promoted Headers:

The first row of the dataset was promoted as the column headers. This allowed each field to be identified correctly and made the dataset easier to work with.

Figure 3: Column headers after promoting the first row

Changed Data Types:

The data types of the columns were changed according to the type of information stored in each field.

Examples include:

- Age → Whole Number
- Room Number → Whole Number
- Billing Amount → Decimal Number
- Date of Admission → Date
- Discharge Date → Date
- Patient ID → Text
- Gender → Text
- Hospital → Text

Correct data types help Power BI perform calculations and analysis accurately.

Capitalized Each Word:

The Capitalize Each Word transformation was applied to the required text fields. This helped maintain a consistent format for text values.

Removed Errors:

Error values were identified and removed from the dataset using the Remove Errors option. This helped improve the quality and reliability of the data.

Trimmed Text:

The Trim transformation was used to remove unnecessary spaces from the beginning and end of text values.

Cleaned Text:

The Clean transformation was applied to remove unwanted non-printing characters from text fields.

Uppercased Text:

The required text values were converted into uppercase where necessary to maintain consistent formatting.

Figure 5: Text cleaning and standardization

Added Index Column:

An Index Column was added to provide sequential numbering for the records. This helps in identifying and organizing individual rows.

Removed Unnecessary Columns:

Columns that were not required for the analysis were removed. This helped reduce unnecessary information and made the data model more organized.

Renamed Columns:

Selected columns were renamed where necessary to make their names clear and meaningful.

Removed Duplicates:

Duplicate records were identified and removed using the Remove Duplicates option. This helped prevent repeated records from affecting the analysis.

AGE GROUP USING CONDITIONAL COLUMN:

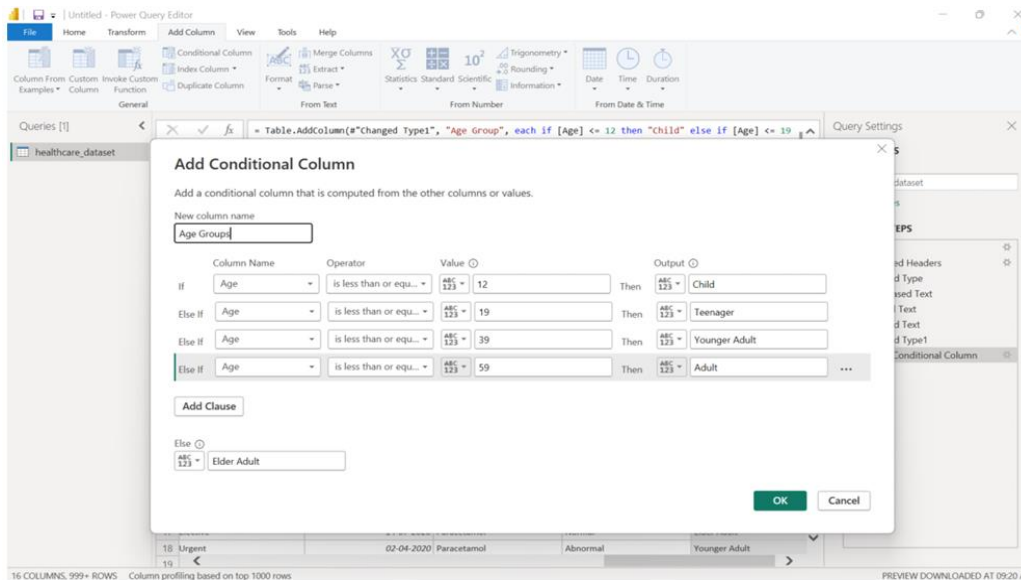
A new column named Age Group was created using the Conditional Column feature in Power Query.

The Age column was used to classify patients into different age categories.

The age groups used are:

- 0-12 → Child
- 13-19 → Teenager
- 20-39 → Younger Adult
- 40-59 → Adult
- 60 and above → Senior Adult

This classification makes it easier to group and analyze patients according to their age.



DATA MODELING:

After completing the data transformation, I organized the healthcare dataset into three separate tables.

The tables created are:

1. Dim_Patient
2. Dim_Admission
3. Fact_Billing

Separating the data into different tables helps maintain a structured model and makes the data easier to analyze.

Dim_Patient:

The Dim_Patient table contains patient demographic information.

The columns include:

- Patient ID

- Name
- Age
- Age Group
- Gender
- Blood Type

This table provides descriptive information about the patients.

Dim_Admission:

The Dim_Admission table contains information related to hospital admissions and medical treatment.

The columns include:

- Admission ID
- Admission Type
- Doctor
- Hospital
- Medical Condition
- Medication
- Room Number
- Test Results

This table helps organize information about hospital admissions and treatments.

Date of Admission	Doctor	Hospital	Room Number	Admission Type	Discharge Date	Medication	Test Res
01 July 2020	Daniel Ferguson	Sons Rich and	316	Elective	14 July 2020	Aspirin	Abnor
04 March 2021	Justin Moore Jr.	Houston PLC	260	Elective	14 March 2021	Aspirin	Abnor
16 May 2023	Matthew Walker	Lindsey Inc	428	Elective	01 June 2023	Aspirin	Abnor
24 March 2024	Kyle Smith	Kemp-Munoz	226	Elective	14 April 2024	Aspirin	Abnor
18 October 2022	Steven James	Luna Inc	227	Elective	11 November 2022	Aspirin	Abnor
17 May 2023	Maria Tran	Thomas-Huber	293	Elective	31 May 2023	Aspirin	Abnor
07 July 2021	Amber Holmes	Brown-Yoder	261	Elective	15 July 2021	Aspirin	Abnor
25 January 2022	Andrea Oneal	Ltd Barnes	438	Elective	28 January 2022	Aspirin	Abnor
06 November 2023	Joshua Singleton	Johnson-Perez	319	Elective	21 November 2023	Aspirin	Abnor
09 July 2021	John Evans	Johnson Goodman Clark, and	283	Elective	04 August 2021	Aspirin	Abnor
30 July 2022	Angela Logan MD	and Woods Liu, Berry	135	Elective	25 August 2022	Aspirin	Abnor
11 January 2021	Angela Reed	Carson and Nielsen, Buchanan	180	Elective	23 January 2021	Aspirin	Abnor
29 May 2019	Michael Miller	Paul Richardson Haynes, and	217	Elective	17 June 2019	Aspirin	Abnor
08 February 2024	Cynthia Stanton MD	Lara-Allen	373	Elective	16 February 2024	Aspirin	Abnor
28 April 2020	Gregory Chung	Poole LLC	283	Elective	20 May 2020	Aspirin	Abnor
30 January 2022	Vickie King	Gentry and Sons	307	Elective	26 February 2022	Aspirin	Abnor
18 June 2019	Peter Wright	Santana Garcia, Alexander and	132	Elective	08 July 2019	Aspirin	Abnor
26 April 2020	Mary Williams	Wells-Price	258	Elective	02 May 2020	Aspirin	Abnor

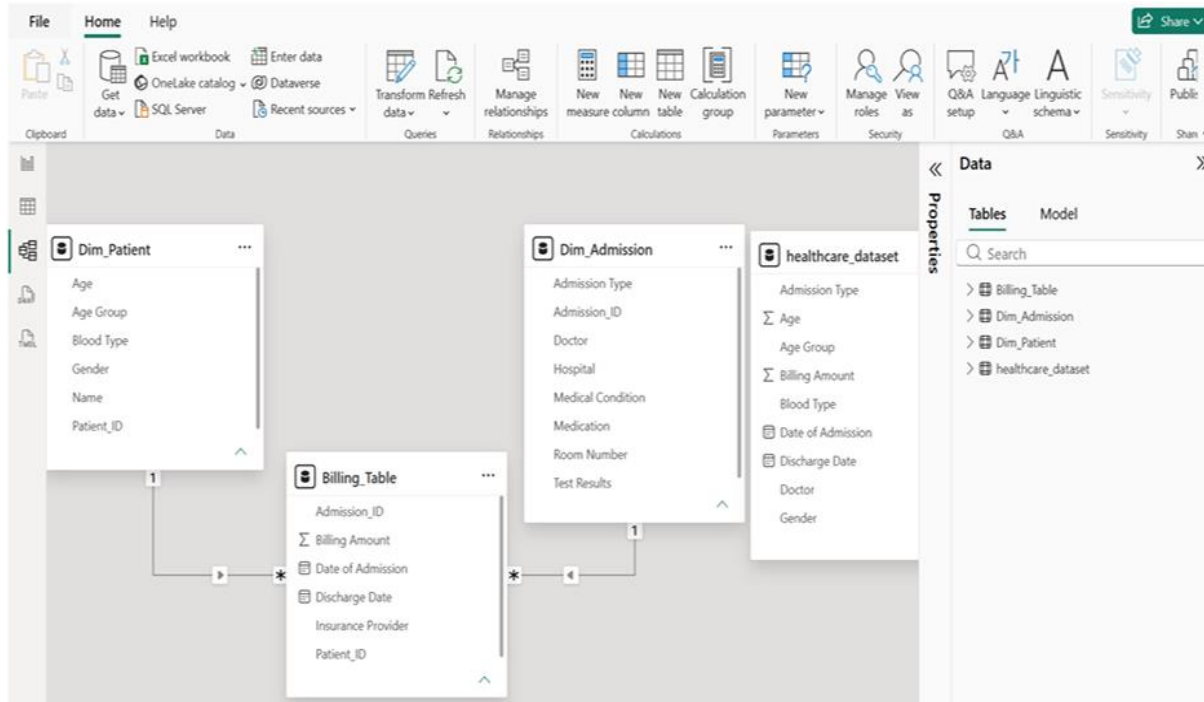
Fact_Billing:

The Fact_Billing table contains billing and admission-related information.

The columns include:

- Patient ID
- Admission ID
- Billing Amount
- Date of Admission
- Discharge Date
- Insurance Provider

This table acts as the main table for future billing and healthcare analysis.



RELATIONSHIPS:

After creating the tables, relationships were established between the tables using their common key fields.

Relationship 1 – Patient and Billing

Fact Billing → Dim Patient

- Common Key: Patient ID
- Cardinality: Many-to-One (*:1)

Relationship 2 – Admission and Billing

Fact_Billing → Dim_Admission

- Common Key: Admission ID
- Cardinality: Many-to-One (*:1)

These relationships connect the tables and allow information from different tables to be used together during analysis.

DATA QUALITY IMPROVEMENTS:

The following improvements were made during the data transformation process:

Data Quality Issue	Action Taken
Incorrect data types	Data types changed
Unwanted spaces	Text trimmed
Non-printing characters	Text cleaned
Inconsistent text format	Text standardized
Error values	Errors removed
Duplicate records	Duplicates removed
Unnecessary information	Columns removed
Age classification	Age Group created
Unclear field names	Columns renamed
Record identification	Index Column added

These transformations improved the quality, consistency and organization of the healthcare dataset.

FINAL DATA MODEL:

The final data model consists of three main tables.

Dim_Patient:

Contains patient demographic information such as Patient ID, Name, Age, Age Group, Gender and Blood Type.

Dim_Admission:

Contains admission and medical information such as Admission ID, Doctor, Hospital, Medical Condition, Medication and Test Results.

Fact_Billing:

Contains billing and date-related information such as Patient ID, Admission ID, Billing Amount, Date of Admission, Discharge Date and Insurance Provider.

The tables are connected through Patient ID and Admission ID.

CONCLUSION:

Various data transformation techniques were applied, including changing data types, trimming and cleaning text, removing errors, standardizing values, removing duplicate records and deleting unnecessary columns. An Age Group column was also created using the Conditional Column feature.

After completing the data transformation, the dataset was organized into Dim_Patient, Dim_Admission and Fact_Billing tables. Relationships were created between the tables using Patient ID and Admission ID with Many-to-One (*:1) cardinality.

This activity helped me gain practical knowledge of Power Query, data cleaning, data transformation, Conditional Columns, data modeling and relationships in Power

BI. The completed data model can be used for further healthcare analysis and visualization.

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